

Exact line search for the convex quadratic function (0.5 pt)

Consider the convex quadratic function

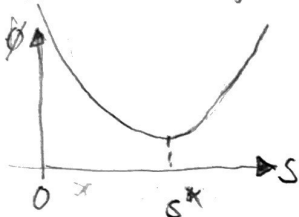
$$f(x) = \frac{1}{2}x^T Qx + q^T x + c,$$

where $x \in \mathbb{R}^n$, and constant parameters Q is a symmetric positive definite matrix $\in \mathbb{R}^{n \times n}$, q is a vector in $\mathbb{R}^n$, and $c \in \mathbb{R}$.

Given a search direction Δx , compute the exact line search parameter t for an arbitrary point x in the domain of f .

Let $\phi(s) = f(x + s\Delta x)$

$$t := \arg \min_{s \geq 0} f(x + s\Delta x) \in \mathbb{R}$$



Solve $\frac{d}{ds} f(x + s\Delta x) = \phi' = 0$ to get t (Note: $D_{\Delta x} f(x) = \frac{1}{\|\Delta x\|} \frac{d}{ds} f(x + s\Delta x) \Big|_{s=0}$)

$$f(x + s\Delta x) = \frac{1}{2}(x + s\Delta x)^T Q(x + s\Delta x) + q^T(x + s\Delta x) + c$$

$$\frac{df}{ds} = (x + s\Delta x)^T Q \Delta x + q^T \Delta x = 0$$

$$\Rightarrow x^T Q \Delta x + s \Delta x^T Q \Delta x + q^T \Delta x = 0$$

$$\Rightarrow s^* = -\frac{q^T \Delta x + x^T Q \Delta x}{\Delta x^T Q \Delta x}$$

Since $s \geq 0$, $t = \max(0, s^*)$

EX5 - Gradient descent with exact line search

Unconstrained minimization problem:

$$\boxed{\text{minimize } \frac{1}{4} x_1^2 + x_2^2 = \frac{1}{2} x^T Q x}$$

$$\nabla f = \begin{bmatrix} \frac{1}{2} x_1 \\ 2x_2 \end{bmatrix} := \Delta x$$

$$\frac{1}{2} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \underbrace{\begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}}_Q \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{4} x_1^2 + x_2^2$$

$$\Delta x^{(0)} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}, x^{(0)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$s^{(0)} = - \frac{x^{(0)T} Q \Delta x^{(0)}}{\Delta x^{(0)T} Q \Delta x^{(0)}} = \frac{\begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \end{bmatrix}}{\begin{bmatrix} -1 & -2 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \end{bmatrix}} = \frac{-1 - 4}{1/2 + 8} = \frac{10}{17}$$

Since $s^{(0)} > 0$, $t^{(0)} = s^{(0)}$

$$x^{(1)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 10/17 \\ -2 \end{bmatrix} = \begin{bmatrix} 24/17 \\ -5/17 \end{bmatrix} \approx \begin{bmatrix} 1.41 \\ -0.18 \end{bmatrix}$$

Before update: $\|\nabla f(x)\|^2 = \frac{1}{4} x_1^2 + 4x_2^2$, $\|\nabla f(x^{(0)})\|^2 = 5$

After " " " " " $\|\nabla f(x^{(1)})\|^2 = \frac{180}{289} \approx 0.623$

→ As expected $\|\nabla f(x)\|^2$ is decreasing

